

Banana fiber reinforced polymer composites for industrial and tribological applications

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16.1 Introduction

Biocomposites reinforced with natural fibers have gained attention in recent times as a promising and sustainable substitute for traditional composites reinforced with synthetic fibers, specifically in applications requiring moderate strength and benign operating environments, such as construction, automotive, and other industrial sectors. Among the various natural fibers, fibers extracted from the stem of plants, known as bast fibers, possess superior strength and stiffness compared to leaf fibers, seed fibers, etc. The commonly used bast fibers include hemp, banana, jute, kenaf, and ramie. Natural fibers, in addition to their biodegradability, are nonabrasive, provide acoustic insulation, and have good thermal properties, making them suitable for use as reinforcement in polymer composites (Chandrasekar et al., 2021).

When solid surfaces move relative to one another, friction and wear are two significant tribological phenomena that typically tend to dissipate energy and damage materials. The topic of interest in this study is the tribological behavior of banana fiber reinforced composites (BFRC). The wear performance of any polymer composite can typically be assessed using the pin-on-disc setup at a laboratory scale (Milosevic et al.,

2020). The tribological behavior of the BFRC has been extensively researched and documented in the literature. Nirmal, in his study, investigated the fiber orientation effects on the wear behavior of the banana/epoxy composite. His study revealed that fiber orientation transverse to the sliding direction provides superior resistance compared to parallel fibers oriented along the sliding direction (Nirmal, 2022). However, research on fiber orientation of the banana/epoxy composite by Chavali and Taru provides contrasting results compared to the former, where fiber orientation changed from parallel (0°) along the sliding direction to the transverse direction (90°) or anti-parallel perpendicular to the sliding direction. This change increases erosion and results in larger material removal, while the friction force increases as the fiber orientation increased from 0° to 90° (Chavali and Taru, 2021). Other such wear studies documented in the literature involve the use of banana fiber along with other natural fibers in hybrid combinations (Krupesh et al., 2023; Thiagarajan et al., 2022; Palanisamy et al., 2022).

Gairola et al. used Taguchi's approach to investigate the influence of wear parameters such as sliding distance, applied load, and sliding velocity (SS) on the tribological behavior of banana/walnut/epoxy composites. Their results revealed that optimum wear performance occurred at 5 wt.% walnut, SS of 1.5 m/s, applied load of 20 N, and sliding distance of 1300 m. The examination of the worn surface under a microscope revealed wear tracks, micro-cracks, and accumulated wear debris in the matrix due to micro-plowing, fiber fracture, fiber-pullout, and fiber-matrix debonding (Gairola et al., 2021). The advantage of Taguchi's approach is that multiple wear parameters can be combined to form a factorial design at three or more levels (Vidhya Sri et al., 2024). Along with Taguchi's orthogonal array, statistical tools such as gray relational analysis (GRA), analysis of variance (ANOVA), 2D or 3D response surface plots or contour plots, and regression analysis have been utilized to obtain the optimum wear parameters for the composites subjected to wear tests (John Iruthaya Raj et al., 2019; Kumar et al., 2024, 2019; Chang et al., 2015). This article also involves determining the optimum wear parameters using methodologies such as GRA and contour plots. GRA is used in optimizing variation within data using sequential and absolute data. In GRA, multiple responses from input parameters are converted into a single output known as gray relational grade (GRG) (Pratap and Patel, 2024). The GRG values close to 1 represent the optimum parameter. 2D and 3D contour plots, plotted as a function of different wear parameters, help in mapping and relating the output with the input conditions. They essentially provide the boundaries and identify zones within which the output can be optimum.

Most of the studies published have reported the influence of wear parameters involving SS, applied load, and sliding distance, while the operating temperature was limited to room temperature. However, increasing the operating temperature during the wear test normally causes increased wear in the polymer composites due to their viscoelastic nature. The higher temperature increases the molecular mobility within the polymer matrix, leading to a decrease in stiffness and accelerating material deformation under such circumstances. Thus, in this study, operating temperature was considered in combination with the wear parameters, and their influence on the wear performance of the banana/epoxy composites was explored.

16.2 Materials and method

16.2.1 Materials

The banana fabric of 200 gsm with a ply thickness of 0.6 mm was purchased from Gogreen Products, Chennai, and LY556 epoxy resin and HY951 hardener were supplied by Herenba Instruments, Chennai. Table 16.1 provides the technical specifications of banana fiber reinforcement and epoxy matrix used in this study.

16.2.2 Fabrication method

The fabrication process is shown in Fig. 16.1. The banana fabric, consisting of 6 layers each of 0.6-mm thickness, epoxy resin, and hardener in a ratio of 10:1, was taken in a weight ratio of 50:50. The fabrics were placed in a mold with dimensions of $300 \times 300 \times 3 \text{ mm}^3$ and then coated with the resin and hardener. The setup was then cured in a hot press at a temperature of 100°C for 2 hours under an applied pressure of 50 psi, followed by curing at room temperature for 24 hours under an applied pressure of 50 psi.

16.2.3 Wear test

The specimens for the wear test were prepared according to the ASTM G99 standard, with a thickness of 3 mm and a length of 32 mm. Fig. 16.2 shows the tribometer setup, which has a steel disc as a counter body with a diameter of 50 mm and a thickness of 10 mm, along with a hardness value of 60 Rockwell hardness. The specimen is fixed to the pin, which presses against the counter body using a specified normal load. As the disc rotates, a wear track forms on its surface due to the sliding action of the pin. The wear tests were conducted under varying applied loads, operating temperatures, and sliding speeds, as designed using the Taguchi L9 orthogonal array, as shown in Table 16.2.

Table 16.1 Specifications of reinforcement and matrix.

Properties	Banana fiber	Epoxy resin
Tensile strength (MPa)	650–780	60–85
Tensile modulus (GPa)	29–32	2.8–3.5
Elongation at break (%)	2–4	1–2
Density (g/cm^3)	1.3	1.1–1.2
Cellulose content (%)	64–65	–
Lignin content (%)	21	–
Fiber diameter (mm)	1–3	–

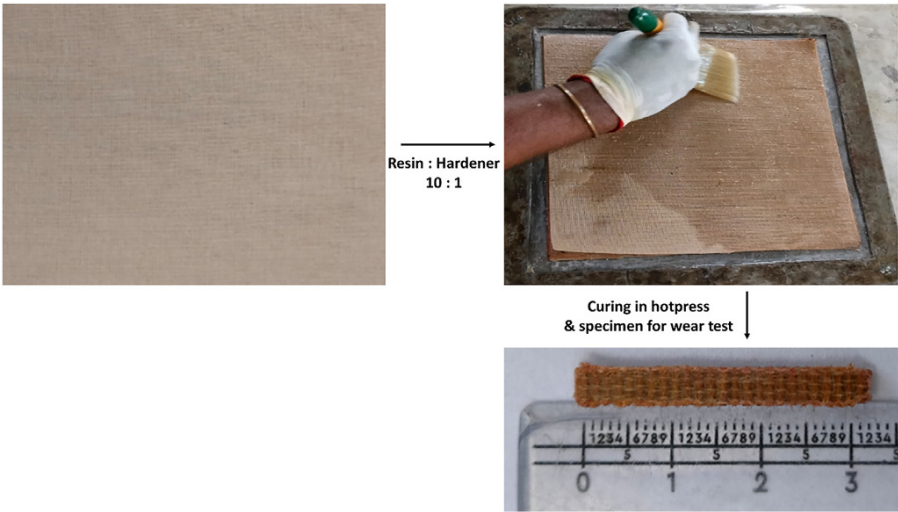


Fig. 16.1 Composite fabrication process and the specimen for the wear test.

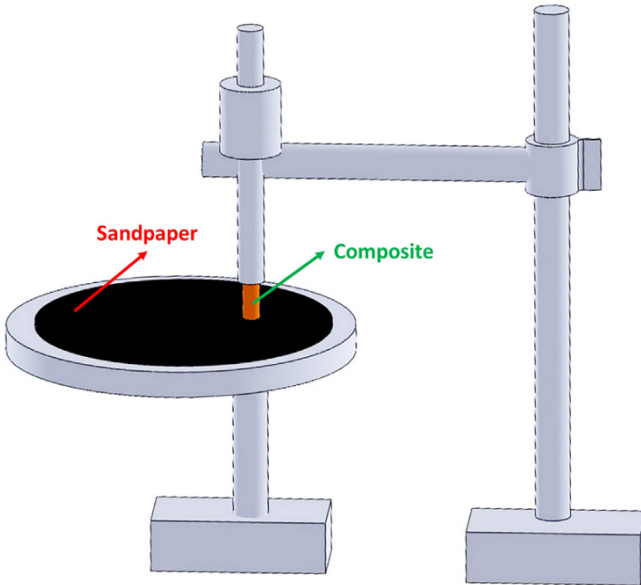


Fig. 16.2 Pin-on-disc test setup.

Table 16.2 Parameters for wear test.

Parameters	Level 1	Level 2	Level 3
Applied load (N)	5	10	20
Sliding speed (rpm)	500	1000	1200
Operating temperature (°C)	30	80	120

The specific wear rate was evaluated using the following mathematical relation:

$$W_s = \frac{\Delta m}{\rho \cdot d \cdot F_N}$$

Where Δm is the difference in mass of the specimen before and after the test, ρ is the density of the composite (kg/m^3), F_N is the normal applied load (N), and d is the sliding distance (m).

The friction coefficient (μ) was calculated by utilizing the following equation:

$$\mu = \frac{\text{Measured friction force}}{\text{Normal applied load}}$$

16.2.4 Morphological analysis of worn surface

The morphological characteristics of the worn surface were examined using the JCM-7000 Benchtop Scanning Electron Microscope at AU Sophisticated Testing and Instrumentation Center, Alliance University, Bengaluru.

16.3 Results and discussion

16.3.1 Wear analysis

Table 16.3 illustrates the specific wear rate (SWR) and coefficient of friction (COF) from the wear test. In order to analyze the influence of wear parameters on the outputs, such as the SWR and COF, contour plots were made, as shown in Figs. 16.3 and 16.4, respectively.

Vidhya Sri et al., 2024 represents the surface plot of SWR with respect to the applied load, operating temperature, and SS. At an applied load of 5 N, SWR was least for higher SS and elevated temperature up to 80°C (Fig. 16.3A) while further increase in the operating temperature up to 120°C was detrimental, as observed from the increase in SWR.

As the applied load was increased to 10 N (Fig. 16.3B), the operating temperature was found to significantly influence the SWR even at lower SS. The contour plot also reveals that SWR can be minimized by increasing the SS. As for the applied load of

Table 16.3 SWR and COF from the wear test.

Data	Applied load in N	Operating temperature (°C)	Sliding speed (rpm)	Mass loss (g)	SWR (m ³ /Nm)	COF (μ)
1	5	30	500	0.0025	3.82263E-13	0.798
2	5	80	1000	0.0023	3.51682E-13	0.629
3	5	120	1200	0.0042	6.42202E-13	0.544
4	10	30	1000	0.0029	2.21713E-13	0.479
5	10	80	1200	0.001	7.64526E-14	0.868
6	10	120	500	0.0053	4.05199E-13	0.218
7	20	30	1200	0.0016	6.11621E-14	0.748
8	20	80	500	0.0029	1.10856E-13	0.431
9	20	120	1000	0.0027	1.03211E-13	0.810

20 N (Fig. 16.3C), both the higher SS and operating temperature aggravate the SWR, thus limiting the banana/epoxy composites to moderate loads for their use in elevated temperatures.

Fig. 16.4A–C depicts the contour plot of the COF with respect to the wear parameters considered. In the case of 5 N, corresponding to Fig. 16.4A, the COF was least for higher SS regardless of the operating temperature. For the applied load of 10 N (Fig. 16.4B and 20 N (Fig. 16.4C), the COF can be kept at a minimum by operating at SS below 700 rpm, while a further increase in both the SS and operating temperature increases the COF.

16.3.2 GRA

Certain rules exist for performing GRA in the machining of composites. The flow of gray relational analysis is described as follows.

Step 1: Normalization/Preprocessing of Data

Here, the original sequence is transformed into a comparable sequence. The experimental data of COF and SWR is normalized between 0 and 1 for the “smaller the better,” “nominal the better,” or “larger the better” criteria. The “smaller the better” criterion was selected, and the data was normalized using Eq. (16.1). The normalized data for the L9 Taguchi array is depicted in Table 16.4.

$$x_i^*(p) = \frac{\max_{\forall i} x_i^o(p) - x_i^o(p)}{\max_{\forall i} x_i^o(p) - \min_{\forall i} x_i^o(p)} \quad (16.1)$$

where $x_i^o(p)$ is the original data sequence value, $x_i^*(p)$ is the normalized sequence value and i is the n th number of results of p th response factor.

Step 2: Evaluating Deviating Sequence

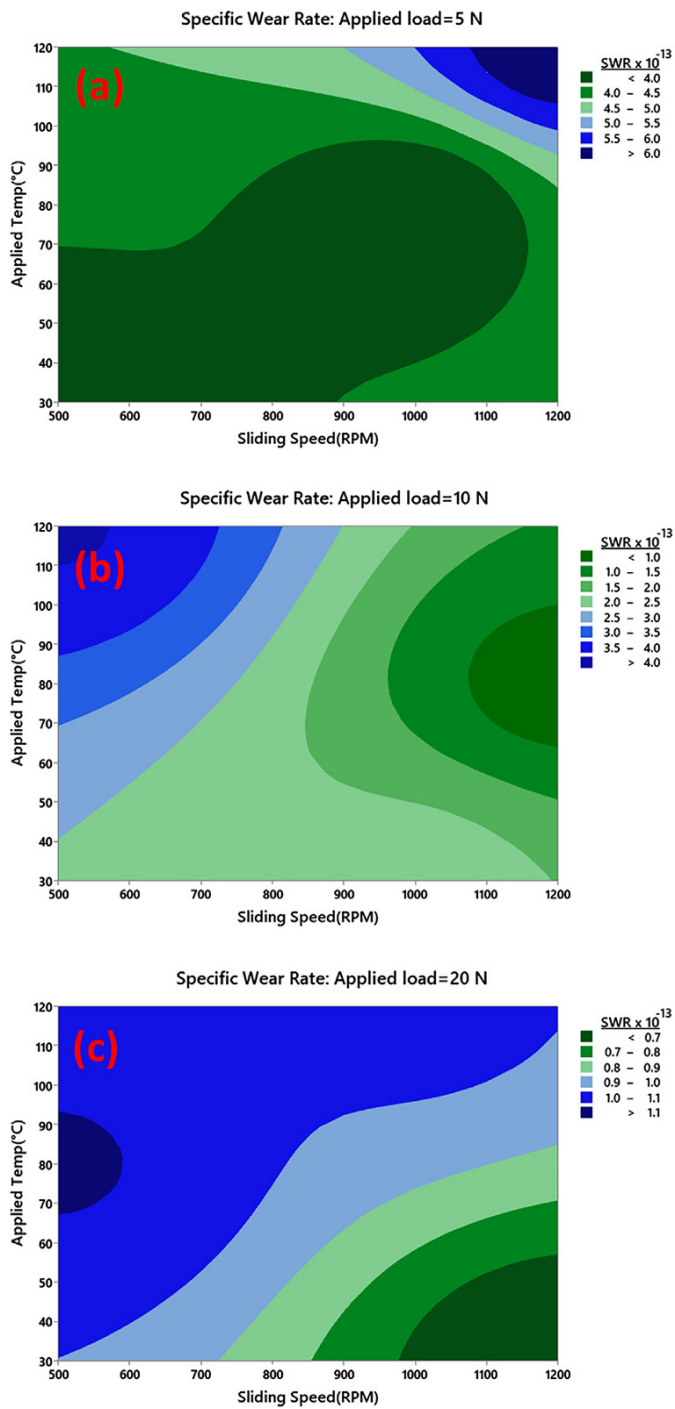


Fig. 16.3 Contour plots for SWR: (A) 5 N, (B) 10 N, and (C) 20 N.

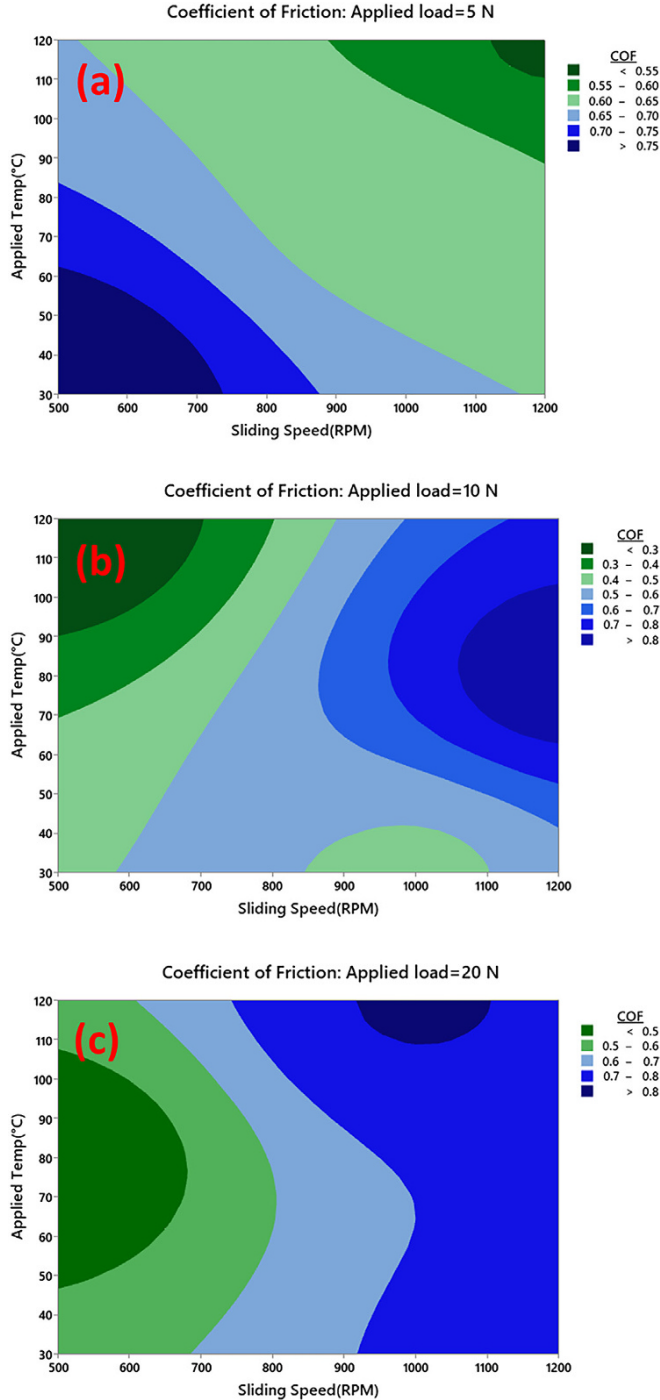


Fig. 16.4 Contour plots for COF: (A) 5 N, (B) 10 N, and (C) 20 N.

Table 16.4 Output from the gray relational analysis.

Data	Normalization		Deviation		Grey relational coefficient		GRG	Rank
	COF	SWR	COF	SWR	COF	SWR		
1	0.108	0.447	0.892	0.553	0.359	0.475	0.417	8
2	0.368	0.500	0.632	0.500	0.442	0.500	0.471	7
3	0.498	0.000	0.502	1.000	0.499	0.333	0.416	9
4	0.598	0.724	0.402	0.276	0.555	0.644	0.599	6
5	0.000	0.974	1.000	0.026	0.333	0.950	0.642	4
6	1.000	0.408	0.000	0.592	1.000	0.458	0.729	2
7	0.185	1.000	0.815	0.000	0.380	1.000	0.690	3
8	0.672	0.914	0.328	0.086	0.604	0.854	0.729	1
9	0.089	0.928	0.911	0.072	0.354	0.874	0.614	5

The deviation sequence was calculated from the normalized data by subtracting the normalized data value from 1. The deviation sequence data results between $\Delta_{\min}$ and $\Delta_{\max}$ in range of 0–1.

Step 3: Evaluating Gray Relational Coefficient (GRC) and Gray Relational Grade (GRG)

GRC was calculated using Eq. (16.2).

$$\xi(p) = \frac{\Delta_{\min} + \zeta \Delta_{\max}}{\Delta_{0i}(p) + \zeta \Delta_{\max}} \quad (16.2)$$

$$\Delta_{0i}(p) = \|x_0^*(p) - x_i^*(p)\| \quad (16.3)$$

$$\Delta_{\min} = \left(\min_{\forall j \in i} \right) \left(\min_{\forall p} \right) \|x_0^*(p) - x_j^*(p)\| \quad (16.4)$$

$$\Delta_{\max} = \left(\max_{\forall j \in i} \right) \left(\max_{\forall p} \right) \|x_0^*(p) - x_j^*(p)\| \quad (16.5)$$

where, Δ_{0i} is the absolute difference between $x_0^*(p)$ and $x_i^*(p)$ and is also called as a deviation sequence. The parameter ζ is the distinguishing coefficient ranging between 0 and 1. In common, ζ value is considered as 0.5 in most of the grey relational analysis.

GRG is the average of each sequence's GRC. It is computed using Eq. (16.6):

$$\gamma_i = \frac{1}{n} \sum_{p=1}^n \xi_i(p) \quad (16.6)$$

However, many factors influence significant outcomes, and they vary accordingly. As a result, Eq. (16.7) can be used to calculate the GRG value.

$$\gamma_i = \frac{1}{n} \sum_{p=1}^n w_p \xi_i(p) \quad (16.7)$$

where w_p is the normalized weight for factor p .

Step 4: Evaluating Rank

Based on GRG, an overall rank was given as tabulated in Table 16.4. The optimum parameter combination based on the GRG value close to 1 is obtained for an applied load of 10 N, spindle speed of 500 rpm, and operating temperature of 80°C.

16.3.3 Signal-to-noise (S/N) ratio and main effect plots

In Taguchi analysis, signal (denoted as “S”) is the desirable value of the output data, whereas noise (denoted as “N”) is the undesirable value of the output data. The S/N ratio is used to measure the quality of the obtained output value deviating from the desired value. Optimum quality with minimum variance is obtained from process parameters with higher S/N ratios. Two types of formulations are available for obtaining the S/N ratios: (1) “Smaller the Better,” where smaller values of the output are desired and (2) “Larger the Better,” where larger values of the output are desired. In the case of the wear behavior of polymer composites, lower values of COF and SWR are desired to improve the wear performance. As a result, “Smaller the Better” was chosen for determining the S/N ratio in this work.

$$\frac{S}{N} = -10 \log_{10} \left(\frac{1}{n} \sum_{k=1}^n \left(\frac{z_k}{\bar{z}} \right)_{ij}^2 \right) \quad (16.8)$$

Here $(z_k)_{ij}$ is the measured response value for i th process variable (COF and SWR) during k th repetition of j th experiment number in Taguchi array (Exp 1, 2, ...); $k = 1, 2, 3 \dots n$ and n is total number of repetitions carried for each experiment in Taguchi array (Exp 1, 2, ...). The S/N ratio for outputs such as COF and SWR is provided in Fig. 16.5A and B. In case of SWR (Fig. 16.4A), optimum wear parameters are obtained at lowest applied load of 5 N and SS of 500 rpm at room temperature while at elevated temperature of 120°C, an applied load of 20 N and SS of 1200 rpm can lower the SWR. For the COF (Fig. 16.5B), higher SS and maximum load of 20 N were required to keep COF to the minimum while operating temperature was detrimental to COF. Increasing the operating temperature accelerates the rise in surface temperature due to friction between the composite and tribometer disc, thereby leading to higher COF at elevated temperatures.

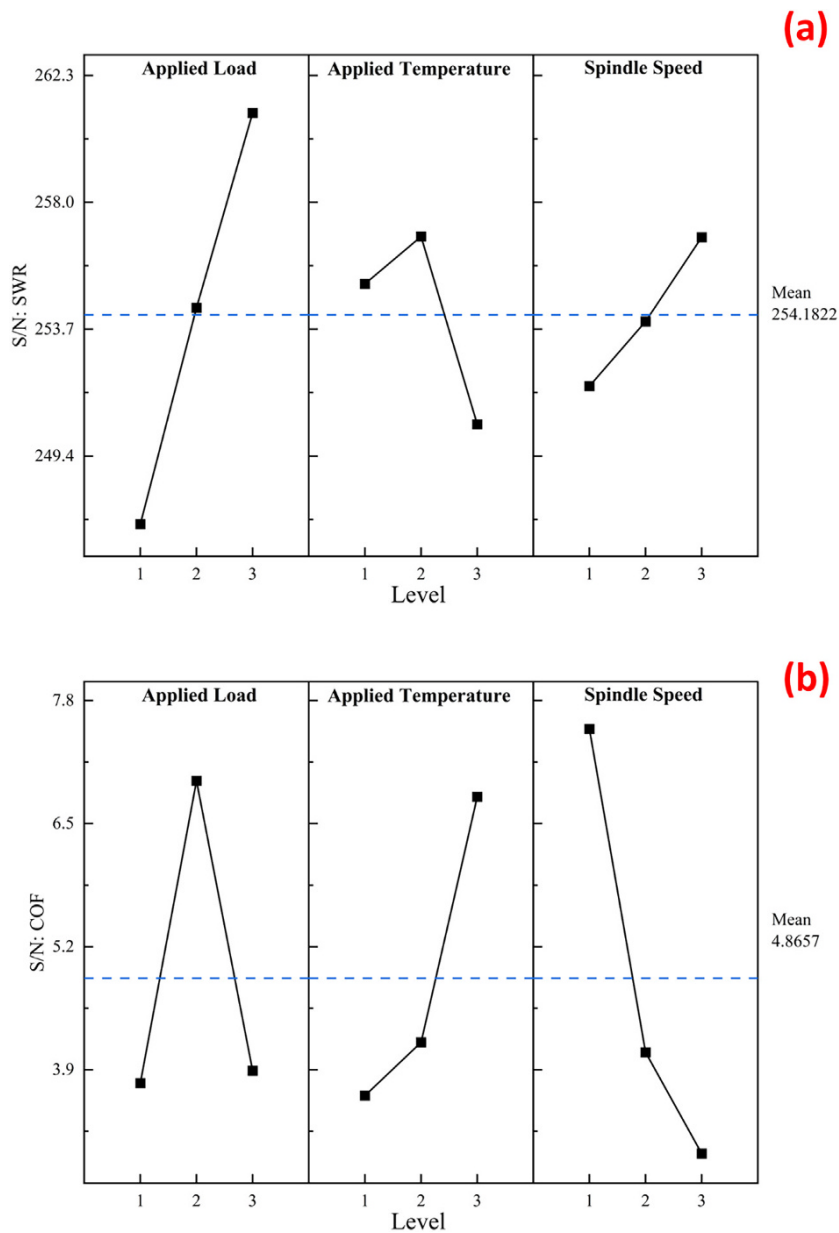


Fig. 16.5 Mean effect plots (A) SWR and (B) COF.

16.3.4 Morphology of the worn surface

The microscopic images of the worn surface of the composites subjected to the wear test are presented in Fig. 16.6A–E. At an applied load of 5 N, both the sliding speeds

of 500 rpm and 1000 rpm, as well as the operating temperature of 80°C (Fig. 16.6A), the worn surface did not exhibit much degradation, with only a few exposed fibers while much of the surface remained intact. As the applied load was increased to 10 N and 20 N, along with a combination of operating temperature and sliding speed, the surface deformation became more prevalent, which can be observed in the form of pits, plowed surfaces, and agglomeration of residual resin particles chipped off from the composite (Fig. 16.6C–E). The residual resin particulates are generated and peel off from the composite surface during sliding as a result of matrix softening caused by the increase in surface temperature due to friction between the composite and the

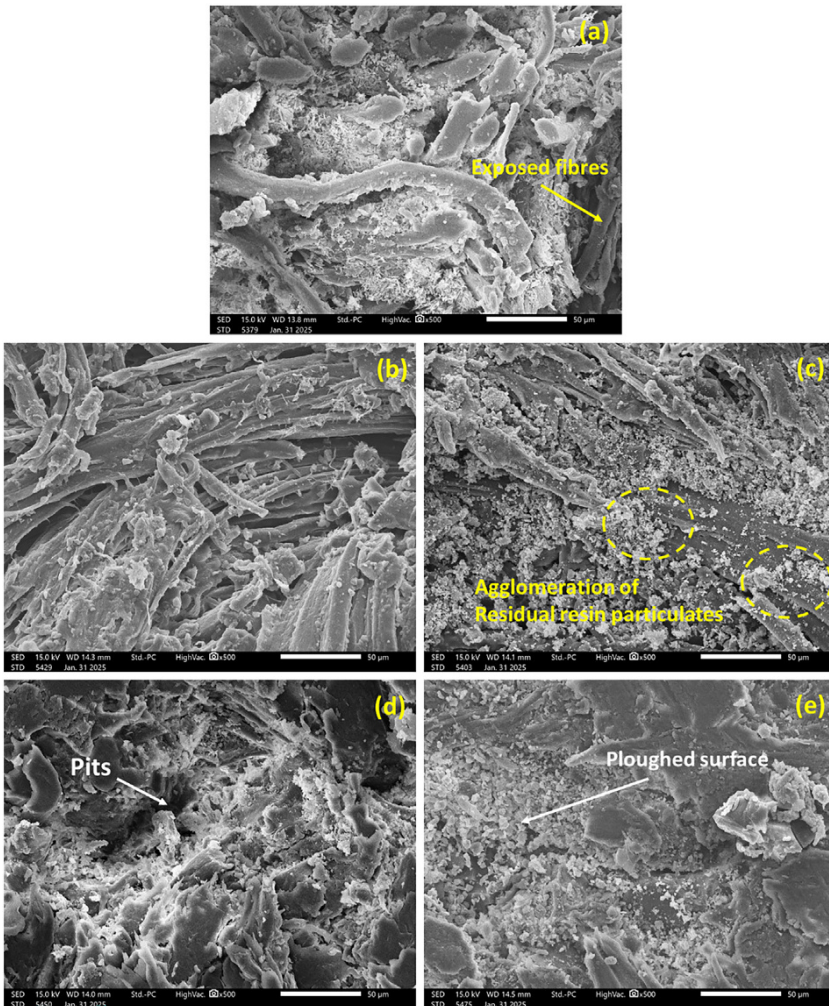


Fig. 16.6 Micrograph of the worn surface of banana/epoxy composites: (A), (C), and (D) at room temperature; (B) and (E) at elevated temperatures.

tribometer disc (Pathan et al., 2016). The peeled-off resin particulates accumulate on the smooth surface of the composite or unaffected zones during the successive sliding distances (Fig. 16.6C).

16.4 Conclusion

The dry sliding wear performance of the banana/epoxy composites was examined in this work based on the applied load, sliding speed, and operating temperature using a pin-on-disc setup. The main finding from this study is that higher sliding speed combined with elevated temperature was detrimental to the wear performance of the composite, as reflected by the higher COF values. The micrographs of the corresponding worn surface also revealed pit formation and agglomeration of residual resin particulates, complementing the accelerated degradation of the surface under such conditions. Thus, in order to obtain optimum performance under wear at elevated temperature, both the sliding speed and applied load have to be nominal.

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